

TS Question Paper 2015**Class 12 Chemistry, Paper - II****Time:3 Hours****Max.Marks:60****Note: Read the following instructions carefully:**

- 1) Answer all the questions of Section-‘A’. Answer any six questions from Section-‘B’ and any two questions from Section-‘C’.**
- 2) In Section-‘A’, Questions from Sr. Nos. 1 to 10 are of “very short answer type”. Each question carries two marks. Every answer may be limited to two to three sentences. Answer all the questions at one place in the same order.**
- 3) In Section-‘B’, Questions from Sr. Nos. 11 to 18 are of “Short answer type”. Each questions carries four marks. Every answer may be limited to 75 words.**
- 4)In Section-‘C’, Questions from Sr.Nos.19 to 21 are of” Long answer type”. Each questions carries eight marks. Every answer may be limited to 300 words.**
- 5)Draw labelled diagram, wherever necessary for questions in section-‘B’ and ‘C’.**

SECTION- A **$10 \times 2 = 20$** **Note: Answer all the questions:****1. Define order of a reaction. Give one example.**

Ans:-The order of reaction with respect to a particular reactant is the exponent to which the concentration term of that reactant, in the rate law is raised. The overall order of the reaction is the sum of the exponents to which the concentration terms in the rate law are raised.

For example:- consider the reaction $aA + bB \rightarrow \text{products}$. The order of the reaction with respect to the reactants A and B is a and b respectively.

The overall order of the reaction is $a + b$.

2. What is the role of cryolite in the metallurgy of Aluminium?

Ans:- Cryolite (Na_3AlF_6) has two roles in the metallurgy of aluminium:

1. To decrease the melting point of the mixture from 2323 K to 1140 K.
2. To increase the electrical conductivity of Al_2O_3 .

3. Why Nitrogen exists as diatomic molecule (N_2) and Phosphorus as P_4 ?

Ans:It is because - N_2 contains a pi bond between two N atoms.

But P_4 does not. It lacks the pi bond and it is because unstable pi tend to overlap among two P atoms.

4. What is “Tailing of Mercury”?

Ans:- Tailing of mercury (a metal) is its innate property. ... It can be removed by adding some impurities in it such as other metal. When ozone is passed through mercury, it loses its meniscus and sticks to the glass due to the formation of mercurous oxide. This is called tailing of mercury.

5. Why $Zn_2 +$ is diamagnetic whereas $Mn_2 +$ is paramagnetic?

Ans:- $Zn_2 +$ is diamagnetic as it has all paired electrons. The valence shell electronic configuration of $Zn_2 +$ is $3d^{10}$.

$Mn_2 +$ is paramagnetic due to presence of five unpaired electrons. The valence shell electronic configuration of $Mn_2 +$ is $3d^5$

6. Calculate molarity of 2.5 grams of Ethanoic Acid(CH_3COOH) in 75 grams of Benzene.

Ans:- Molar mass of ethanoic acid = 60

Molar mass of benzene = 78

No. of moles of ethanoic acid = $2.5/60 = 0.041$

No. of moles of benzene = $75/78 = 0.96153$

Mole fraction = $0.041/(0.041+0.96153) = 0.04$

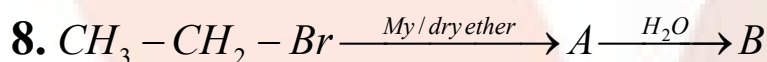
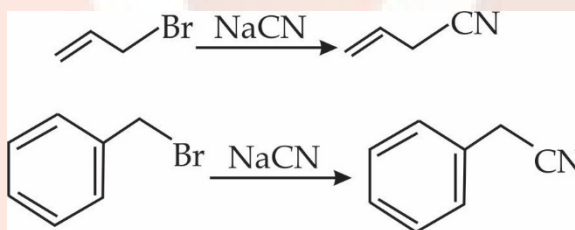
Molality = No. of moles of solute/weight of the solvent (in kg)

$$= 0.041/0.075$$

$$= 0.5466\text{M}$$

7. Explain $\text{S}_\text{N}2$ reaction with one example.

Ans: The $\text{S}_\text{N}2$ reaction is a nucleophilic substitution reaction where a bond is broken and another is formed synchronously. Two reacting species are involved in the rate determining step of the reaction. The term ' $\text{S}_\text{N}2$ ' stands for – Substitution Nucleophilic Bimolecular. This type of reaction is also referred to as bimolecular nucleophilic substitution, associative substitution, and interchange mechanism.



Identify A and B compounds.

Ans:- A - $\text{CH}_3-\text{CH}_2-\text{MgBr}$

B- $\text{CH}_3\text{CH}_3 + \text{MgBrOH}$

9. Explain Carbylamine reaction with one example.

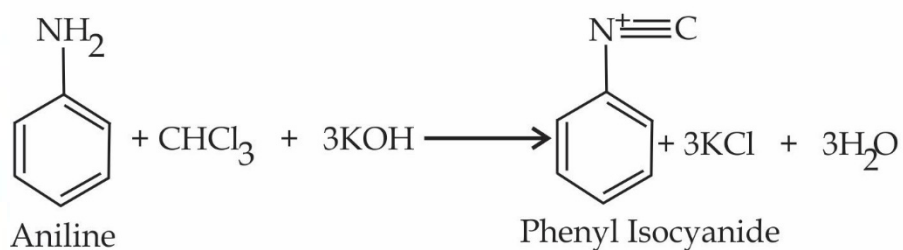
Ans:- By the help of reaction of a primary amine, chloroform and base an isocyanide is synthesized, this is known as

carbylamine reaction.

carbylamine reaction written as:



example:-



10. Arrange the following in decreasing order of their basic strength:



Ans:- $(C_2H_5)_2NH \rightarrow C_2H_5NH_2 \rightarrow NH_3 \rightarrow C_6H_5NH_2$

SECTION – B

(6 × 4 = 24)

Note : Answer any six questions.

11. Derive Bragg's equation.

Ans:-The phases of the beams coincide when the incident angle equals reflecting angle. The rays of the incident beam are in phase and parallel upto point z, which is the point at which top beam strikes the top layer. The second beam passes to next layer and is scattered by B. The second beam travels extra distance $AB + BC$. This extra distance is an integral multiple of the wavelength.

$$n\lambda = AB + BC.$$

But

$$AB = BC$$

$$n\lambda = 2AB \dots (1)$$

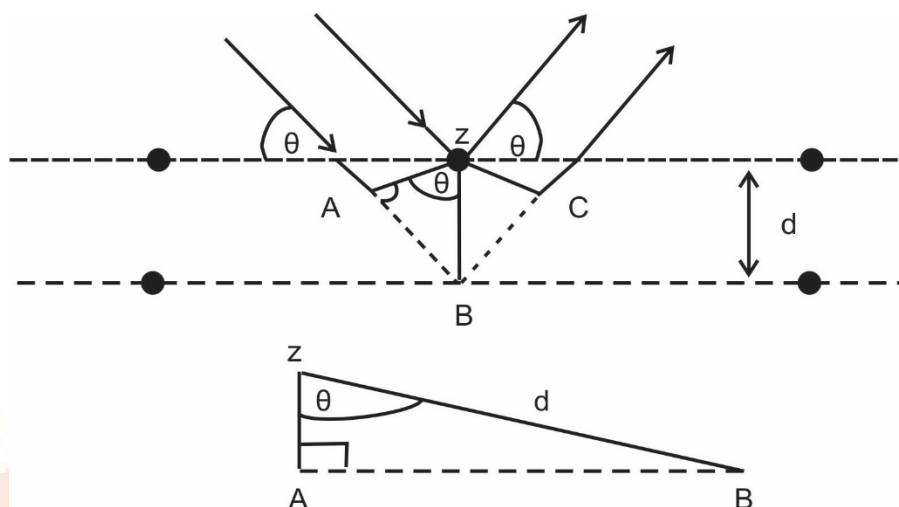
d is the hypotenuse of the right triangle Abz. Ab is opposite to angle θ

$$AB = d \sin \theta \dots (2)$$

Substitute equation (2) in equation (1)

$$n\lambda = 2d \sin \theta$$

This is equation for Bragg's law



12. Give any four differences between Physiosorption and Chemisorption.

Ans- Physiosorption:-

1. No new compound is formed in the process.
2. It is favoured by low temperature conditions.
3. It is an example of multi-layer adsorption
4. In this type of adsorption, the adsorbate is attached to the surface of the adsorbent with weak van der Waal's forces of attraction.

Chemisorption:-

1. New compounds are formed at the surface of the adsorbent.
2. It is favoured by high temperature conditions.
3. It is an example of mono-layer adsorption.

4. In this type of adsorption, strong chemical bonds are formed between the adsorbate and the surface of the adsorbent.

13. Explain the purification of Sulphide ores by Froth Flotation method.

Ans:- Sulphide ores are separated by the froth flotation method in which a mixture of water and pine oil which is made to froth in a tank.

- It is first taken in the tank. A compressed air is blown through the pipe which is used to rotate the agitator to produce the froth.
- Sulphide ore is first wetted and then it is coated with the pine oil and drawn into the agitator to produce the froth.
- Sulphide being more negative which attracts the oil and the gangue becomes less negative which is attracted by the water.
- Froth containing the sulphide ore is transferred to another container, washed and dried.

14. Explain Werner's theory of coordinate compounds with suitable examples.

Ans:- Werner's theory of co-ordination compounds proposed that a central metal atom or ion shows two different types of valency, namely primary valency and secondary valency. The primary valency is equal to the oxidation state of the central metal atom or ion and secondary valency equals the

coordination number of the central metal ion.

The coordination number of a central metal atom or ion is always fixed for a particular oxidation state.

According to Werner's coordination theory, the central metal atom present in the complex can exhibit two types of valencies which are given below:

(i) Primary Valency (Ionizable Valency):

- Primary Valency of a central metal atom is equal to the oxidation state of the central metal atom.
- The primary Valency of the central metal atom can be satisfied by negative species.
- Primary Valency exhibiting species being non-directional in nature.

(ii) Secondary Valency (Non-Ionizable Valency) :

- Secondary valency of the central metal atom is equal to the coordination number of the central metal atom.
- The secondary valency can be satisfied by negative, positive, or neutral species.
- Secondary valency exhibiting the species being directional in nature.
- The Secondary valency is responsible for the geometry of the complex as:
- Secondary valency exhibiting species can retain their individual identity

15. Give the sources of the following vitamins and name the diseases caused by their deficiency:

- a) A b) D c) E d) K

Ans:- Vitamins Sources Diseases caused by deficiency

- a) A Apricots liver disorders, hematopoiesis
b) D Cheese, Liver soft bones, skeletal deformities
c) E Nuts, Seeds Disorientation and vision problem
d) K Fish, Eggs Bleeding, hemorrhagic disease of the newborn.

16. Write short notes on Antiseptics and Disinfectants.

Ans:- Antiseptics:-

These drugs are applied to living tissues to kill the bacteria and to stop their growth in wound, thus preventing its infection. Antiseptics do not heal the wound but prevent the wound from getting infected. Dettol (a mixture of terpineol and chloroxylenol) is an antiseptic. Other antiseptics are tincture iodine, bithional, iodoform, boric acid, phenol (dilute solution) etc.

Disinfectants:-

Disinfectants are applied to non-living objects to kill the microorganisms, used in public health sanitation, floors, to streamline instruments, etc. Examples include chlorine, sulphur dioxide, phenol (concentrated solution).

17. a) What is addition polymer? Give example.

Ans: An addition polymer is formed by the combination of several monomers that bind together via rearrangement of bonds under specific conditions of pressure heat and/or presence of catalysts. In a complete polymer, the double bonds are converted into single bonds without the loss of any atom or molecule.

Example:-PVC, Polystyrene

b)What is PHBV? How is it useful to man?

Ans:- PHBV is a copolymer. It is polyhydroxy butyrate -CO- β -hydroxy valerate. The monomers used are 3-hydroxy butanoic acid and 3- hydroxy pentanoic acid. It has an ester linkage. It is useful to mankind because it is biodegradable polymer and thus prevents environmental pollution.

18. What is relative lowering of vapour pressure? How is it useful to determine the molar mass of a solute?

Ans:- Vapour pressure of a solvent present in solution is less than the vapour pressure of the pure solvent.

$$p_1 = x_1 p_{10}$$

The reduction in the vapour pressure of solvent (p_1) is given as:

$$\begin{aligned}\Delta p_1 &= p_1^0 - p_1 = p_1^0 - p_1^0 x_1 \\ &= p_1^0 (1 - x_1)\end{aligned}$$

As $x_2 = 1 - x_1$ the above equation reduces to $\Delta p_1 = X_2 - p_1^0$

$$\frac{\Delta p_1}{p_1^0} = \frac{p_1^0 - p_1}{p_1^0} = x_2$$

$$\frac{p_1^0 - p_1}{p_1^0} \text{ relative lowering of vapour pressure}$$

$$x_2 \text{ mole fraction of the solute} = \frac{n_2}{n_1 + n_2}$$

Where n_1 and n_2 are the number of moles of solvent and solute respectively present in the solution.

The above equation can be written as:

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{n_2}{n_1 + n_2}$$

For dilute solutions $n_2 \ll n_1$, hence ignoring n_2 in the denominator we get

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{n_2}{n_1}$$

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 \times m_1}{M_2 \times w_1}$$

According to Raoult's law the relative lowering of vapor pressure is,

$$\frac{p_{\text{solvent}}^0 - p_{\text{solution}}}{p_{\text{solvent}}^0} = X_B$$

Let w_A and w_B be the weights of the solvent and solute respectively and their corresponding molar masses are M_A and M_B , then the mole fraction of the solute x_B is

$$X_B = \frac{n_B}{n_A + n_B} \quad (9.20)$$

Here, n_A & n_B are the moles of the solvent and the solute respectively. For dilute solutions $n_A \gg n_B$. Hence $n_A + n_B \approx n_A$.
Now

$$X_B = \frac{n_B}{n_A}$$

Number of moles of solvent and the solute are,

$$n_A = \frac{w_A}{M_A}, n_B = \frac{w_B}{M_B} \quad (9.21)$$

$$\text{Therefore, } X_B = \frac{\frac{w_B}{M_B}}{\frac{w_A}{M_A}}$$

Thus

$$\text{Relative lowering of Vapor pressure} = \frac{\frac{w_B}{M_B}}{\frac{w_A}{M_A}}$$

$$\text{Relative lowering of Vapor pressure} = \frac{w_B \times M_A}{w_A \times M_B}$$

..... (9.22)

From the equation (9.35) the molar mass of the solute (M_B) can be calculated using the known the values of w_A , w_B , M_A and the measured relative lowering of vapour pressure.



SECTION – C

(2 × 18 = 16)

Note : Answer ANY TWO questions.

19. a) Explain the effect of temperature and catalyst on the rate of a reaction.

Ans:- For a reaction to occur between particles it needs activation energy. A catalyst provide an alternative pathway for the reaction, that has a lower activation energy. Therefore more the particles with the activation energy will increase the rate.

Increasing the temperature has no effect on the activation energy. The temperature changes the amount of energy the particles have, therefore increasing the temperature mean there are more particles with high energy greater than the activation energy. This means that an increase in temperature will increase the rate.

b) State Faraday's First Law of Electrolysis.

Ans:- Faraday's First Law of Electrolysis states that the chemical deposition due to the flow of current through an electrolyte is directly proportional to the quantity of electricity (coulombs) passed through it.

i.e. mass of chemical deposition:

$$m \propto \text{Quantity of electricity}, \Rightarrow m = Z.Q$$

Where, Z is a constant of proportionality and is known as electro-chemical equivalent of the substance.

20. How is Chlorine prepared in the laboratory? Explain how it react with following along equations.

a) Hot Concc. NaOH

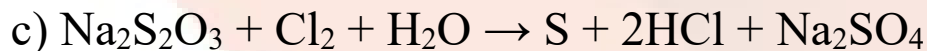
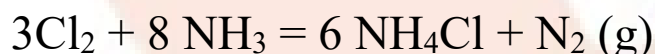
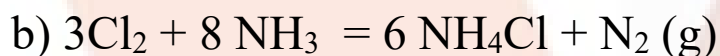
b) NH₃(Excess)

c) Na₂S₂O₃

Ans:- Chlorine is prepared by using manganese dioxide to oxidised hydrochloric acid,



The manganese dioxide is heated with concentrated HCl and the gas produced is bubbled through water to remove HCl gas present.



21. Explain the following with one example:

a) Aldol condensation

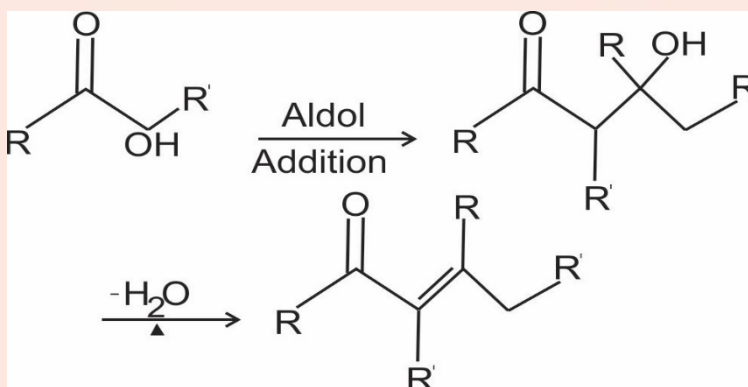
b) Cannizzaro reaction

c) Esterification

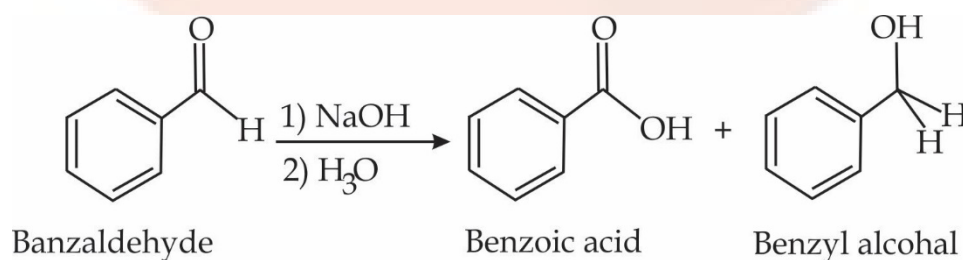
d) Decarboxylation

Ans:-

a) Aldol condensation :- the adducts obtained from the Aldol Addition can easily be converted (in situ) to α,β -unsaturated carbonyl compounds, either thermally or under acidic or basic catalysis. The formation of the conjugated system is the driving force for this spontaneous dehydration. Under a variety of protocols, the condensation product can be obtained directly without isolation of the aldol.



b) Cannizzaro reaction:- Cannizzaro reaction is a reaction in which two molecules of an aldehyde are reacted to produce a primary alcohol and a carboxylic acid using a hydroxide base. When two molecules of aldehyde is treated with sodium hydroxide, a reaction occurs in which one mole of aldehyde is oxidized (giving carboxylic acid) while the other mole of aldehyde is reduced (giving primary alcohol)

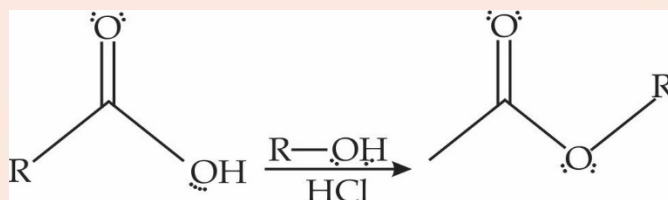


c) Esterification :- Esterification is when two reactants basically

form an ester in the end. But you knew that.

A common one is called the Fischer esterification, which is when excess/xs alcohol reacts with a carboxylic acid in (other) acid.

Here is an example of a general carboxylic acid reacting with a general alcohol in HCl



d)Decarboxylation:-Decarboxylation reaction is defined as a chemical reaction that eliminates a carboxyl group and liberates carbon dioxide (CO_2). Decarboxylation mostly refers to a reaction of carboxylic acids erasing a carbon atom from a chain of carbons.

For, Ex: R-COOH (a carboxylic acid) $\rightarrow \text{R-H} + \text{CO}_2$

or, R-COO-Na (sod. salt of the acid) $+ \text{NaOH} \rightarrow \text{R-H} + \text{Na}_2\text{CO}_3$